SHAMIR KHAN

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EDUCATION

Bachelor of Science in Computer Science — **Expected April 2028**
Wilfrid Laurier University — *Waterloo, ON*
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (OOP), Distributed Systems, Operating Systems & Concurrency, Database Systems, Computer Networks, Systems Architecture, Discrete Mathematics

FEATURED ENGINEERING PROJECTS

CloudGuard – Distributed Cloud Security Posture Platform — **July 2026**
C++17, Kafka, Scala Spark, Java Spring Boot, Python ML, PostgreSQL — GitHub / Live Demo

- **Architected an event-driven distributed microservices platform** across 5 containerized services connected via **Apache Kafka**, achieving **99.9% uptime** and handling multi-cloud telemetry ingestion across AWS, Azure, and GCP.
- **Engineered a native C++ telemetry agent** using Windows kernel APIs (`GlobalMemoryStatusEx`, `GetSystemTimes`) and POSIX parsers to stream machine metrics to Kafka topics at sub-millisecond overhead.
- **Built an unsupervised scikit-learn IsolationForest ML pipeline** scoring live Kafka streams in **507ms (p50)**, detecting SSH brute-force intrusions and billing spikes to cut incident response MTTR by **85%**.
- **Configured automated GitHub Actions CI/CD gates** running unit and integration test suites with 100% pass rate.

NexusQueue – Distributed Asynchronous Task & DAG Engine — **Nov 2026**
TypeScript, Node.js 22, WebSockets, Redis Streams, Docker — GitHub / Live Demo

- **Engineered a distributed background task engine** benchmarked at **119,000 jobs/sec** enqueue throughput and **3,418 tasks/sec** execution across 50 concurrent worker pipelines with strict at-least-once delivery guarantees.
- **Implemented a Binary Min-Heap Priority Queue** with `O(log n)` scheduling and FIFO tie-breaking, combined with distributed **Token Bucket** rate limiters and AWS Full Jitter backoff against thundering-herd storms.
- **Architected a DAG Workflow Scheduler** resolving multi-stage task dependency trees using **Kahn's topological sort**, paired with an automated **Zombie Worker Watchdog** that reclaims orphaned tasks during node crashes.
- **Deployed a real-time WebSocket dashboard** streaming live `p50, p90, p95, p99` latency percentiles and Prometheus metrics.

LedgerMesh – High-Concurrency Double-Entry Ledger Core — **Nov 2026**
Python 3.13, FastAPI, Concurrency (OCC), Merkle Trees, Prometheus — GitHub / Live Demo

- **Architected an institutional clearing engine** benchmarked at **10,790 tx/sec** with a median (p50) latency of **0.05ms**, enforcing mathematical double-entry invariants (Σ Debits = Σ Credits) and zero floating-point drift.
- **Designed Optimistic Concurrency Control (OCC)** with account-level versioning and exponential backoff, eliminating race conditions and double-spending across 20 concurrent worker coroutines without database locks.
- **Implemented a cryptographic SHA-256 Merkle audit trail** linking sequential transaction blocks, providing 100% tamper-evident integrity proofs and automated reconciliation against external bank statements.
- **Authored 28 automated test vectors** (100% pass rate) and integrated Prometheus endpoints for live latency tracking.

AlphaPulse – Quantitative Algorithmic Trading Terminal — **Sept 2026**
FastAPI, NumPy, Pandas, OpenCV, Gemini 3.8-Flash — GitHub / Live Demo

- **Built a quantitative execution terminal** achieving **245,747 bars/sec** feature ingestion throughput across 6 technical indicators (EMA ribbons, Supertrend, ATR, MACD) via vectorized NumPy computations.
- **Engineered a 1-year comparative multi-strategy backtester** evaluating Sharpe (6.39), Sortino (9.68), and Calmar ratios, demonstrating a **44% reduction in maximum drawdown** (8.93% vs 15.96%) over passive buy-and-hold.
- **Constructed a dual-mode Computer Vision pipeline** using OpenCV heuristics and multimodal **Gemini 3.8-Flash AI** to analyze clipboard chart screenshots (`Ctrl+V`) with sub-second response times.

TECHNICAL EXPERIENCE

Lead Web Developer & Systems Administrator — **Jan 2026 – Present**
Wilfrid Laurier University Student Association (WLU MSA) — *Waterloo, ON* | *wlumsa.org*

- **Architected and deployed a production web platform** serving **250+ active campus members** using **Next.js**, **React**, **TypeScript**, and **Tailwind CSS**, achieving 99.9% uptime.
- **Engineered dynamic donation and event modules** integrated with **Payload CMS** and payment webhooks, supporting student initiatives that processed **$50K+** in charitable contributions.

Operations and Technical Lead — **August 2026**
The J Healthcare Initiative (via Riipen Labs) — *Remote, ON*

- **Led technical discovery and digital audits** for a healthcare non-profit, identifying UX bottlenecks and authoring a 30–60–90 day technical execution roadmap to restructure resource funnels to under 3 clicks.

TECHNICAL SKILLS SUMMARY

Languages: C++, Python, TypeScript, Java, JavaScript, Scala, Kotlin, SQL, HTML5/CSS3, C

Distributed Systems & Cloud: Apache Kafka, Apache Spark, Redis Streams, Docker, WebSockets, GitHub Actions (CI/CD), Linux/Unix, Render, Vercel, AWS

Frameworks & Databases: FastAPI, Node.js, Express, React, Next.js, Spring Boot, PostgreSQL, MongoDB, SQLite, NumPy, Pandas, OpenCV, Tailwind CSS

Algorithms & Core Concepts: Optimistic Concurrency Control (OCC), Binary Min-Heaps, DAG Scheduling (Kahn's Algorithm), SHA-256 Merkle Trees, Token Bucket Rate Limiting, AWS Full Jitter Backoff, Object-Oriented Programming (OOP), Test-Driven Development (TDD), RESTful APIs